(2) Magnetic field and stationary currents

Historically magnetism was discovered by the greeks, also described by Thales.

Magnesian stone : Magnetis lithos

Interaction of magnets resembles to that of charges, they were looking for the magnetic charge or monopole.

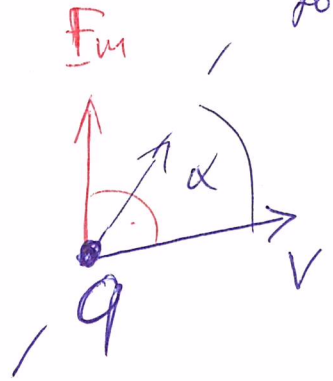
Hans Christian Oersted (1820):

Connects magnetic force to currents!

Currents ⟶ magnetic field

Magnetic field ⟶ force on moving charge (Hendrik Anton Lorentz 1891)

force-free line



$$F_m \sim q v \sin\alpha$$

$$F_m = B q v \sin\alpha$$

$$\boxed{\underline{F}_m = q\,\underline{v} \times \underline{B}}$$

$\underline{B}$: magnetic induction vector

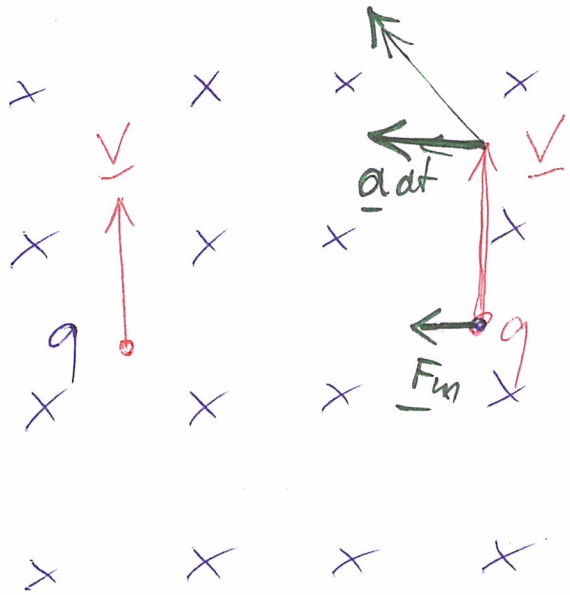
dimensions: $N = A\,s \cdot \frac{m}{s} \cdot [B] \Rightarrow [B] = \frac{1N}{Am} = 1T$

Induction lines are not pointing towards the direction of force, but to the force-free direction.

Induction lines always form closed loops!

Motion of a charged particle in homogenous (uniform) magnetic field

suppose $\underline{v} \perp \underline{B}$



$$\underline{F} = q\,\underline{v} \times \underline{B}$$

$$m\underline{a} = q\,\underline{v} \times \underline{B}$$

$$\underline{a} = \frac{q}{m} \cdot \underline{v} \times \underline{B}$$

$\Downarrow$ $\quad \underline{a} \perp \underline{v}$

result: circular motion

$$\underline{a}_{cp} = \frac{v^2}{R}$$

$$\frac{v^2}{R} = \frac{q}{m}|\underline{v} \times \underline{B}| \Rightarrow \quad R = \frac{m v^{\cancel{2}}}{q \cancel{v} B} = \frac{m v}{q B}$$

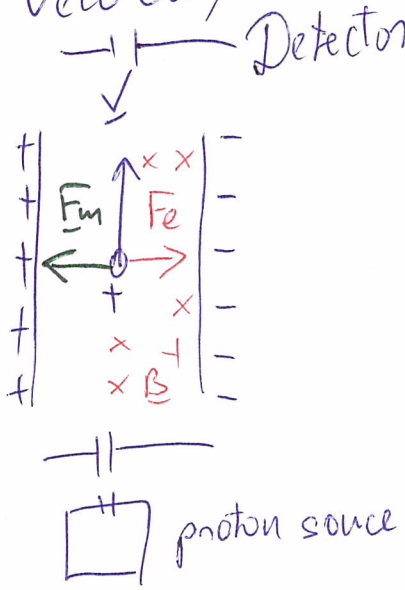
by measuring R one can measure q, m, ω, v

$$\omega = \frac{v}{R} = \frac{\cancel{v}}{m \cancel{v}} \cdot q\cancel{B} = \frac{qB}{m} \quad \text{independent of } v, R$$

cyclotron frequency

Simple applications :

(1) Velocity selector :

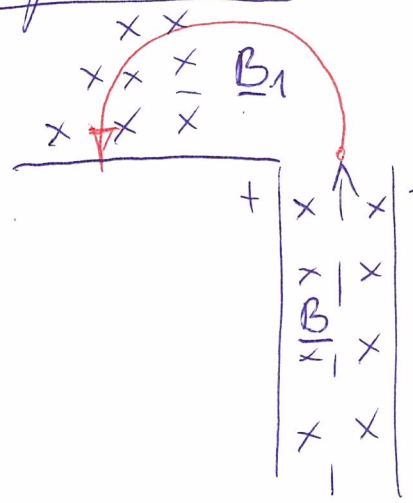


$$\underline{F}_{TOT} = q\underline{E} + q\underline{v} \times \underline{B}$$

$$\underline{F}_{TOT} = 0 \rightarrow qE = qvB$$

$$v = \frac{B}{E}$$

(2) Mass spectrometer :

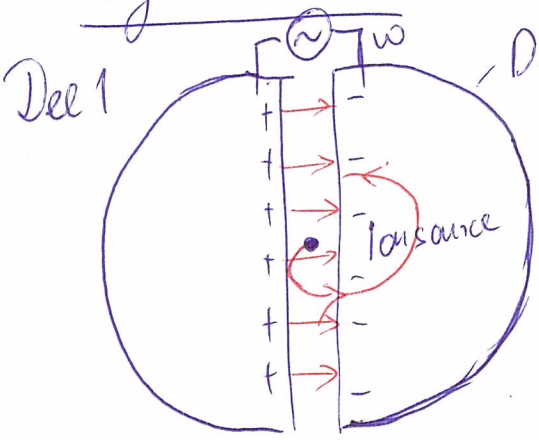


$$R = \frac{mv}{qB_1} \Rightarrow \frac{m}{q} = \frac{RB_1}{V} = \frac{R \cdot B_1 B}{E}$$

Knowing that ions have low integer multiple of elementary charge, we can measure the mass of the ions.

discovery of e^- by Thomson.

(3) Cyclotron :



$$\omega_c = \frac{qB}{m} \quad \text{constant!}$$

$$R = \frac{mv}{qB}$$ the larger the size and B the larger the speed or kinetic energy.

$$E_{kin} = \frac{1}{2} m v^2 = \frac{q^2 B^2 R^2}{2m}$$

Can one grow energy infinitely? NO!

when approaching c ~~one~~ ω_c is reduced!

Maximum with cyclotron $\sim$ 20 MeV

(eV as energy unit: the energy of an object having a charge of an electron (or proton) accelerated by 1V

$$E_{pot} = q \, E \cdot l = q \, \Delta U$$

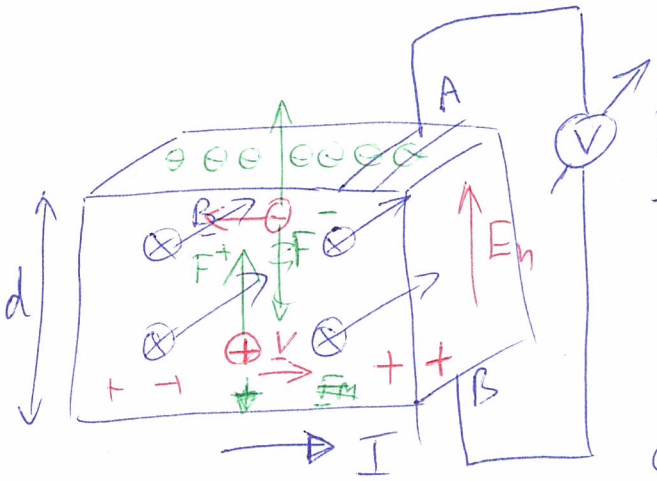
$$1 J = 1 C \cdot V$$

$$1 eV = 1{,}6 \cdot 10^{-19} C \cdot 1 V = 1{,}6 \cdot 10^{-19} J$$

Synchrotron: Both B and ω_c tuned to remain at a constant R.

(Largest: LHC R = 4,3 km, 6,8 TeV)

What happens if a current carrying conductor is placed in magnetic field?



$$\underline{F}_m^+ = q\,\underline{v}_+ \times \underline{B} = \underline{F}^-$$

with a volt-meter you can determine the sign of the charge carriers!

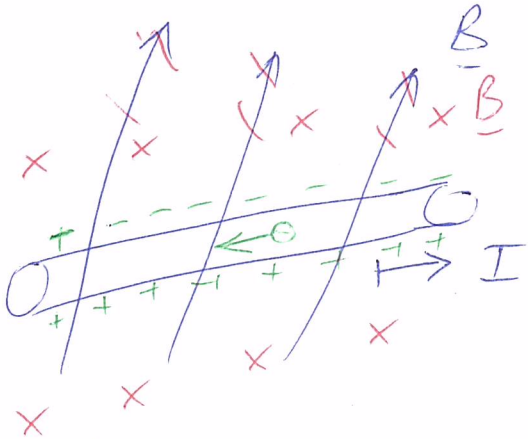
$$F_m^- = q\,v_d \cdot B = q\,E_H$$

$$\Delta U^A(B)_{Hall} = v_d \cdot B \cdot d$$

One can measure the v_d drift velocity.

Knowing $I = v_d\, n\, q\, A$ one can measure n or if knowing n one can measure B by a simple volt-meter! Hall ~~send~~ sensor (B)

Does it acts on the wire itself?



slightly displaced - charges will attract the remaining + charge in the crystal

↳ The force is transferred to the wire!

If the charge is carried by ⊕ charges, the force is the same!

$$\vec{F}_m = q\underline{v} \times \underline{B}$$

$\underline{F}_m = q\,\underline{v} \times \underline{B}$ for a single e^-

the number of e^--s in a dr length:

$$dN = n \cdot A \cdot dr$$ (dr is so small, that $\underline{B}$ and $\underline{v}$ const.

$$d\underline{F} = dN\,\underline{F}_m = qnA\,\underline{v} \times \underline{B}\,dr = A \cdot \underline{j} \times \underline{B}\,dr =$$

$$d\underline{F} = I \cdot (\underline{dl}_j \times \underline{B})\,dr \qquad \underline{u}_j : \text{unit vector} \parallel \underline{j}$$

$$\underline{F} = I \int_1^2 (\underline{dl}_j \times \underline{B})\,dr$$

for a straight conducting wire:

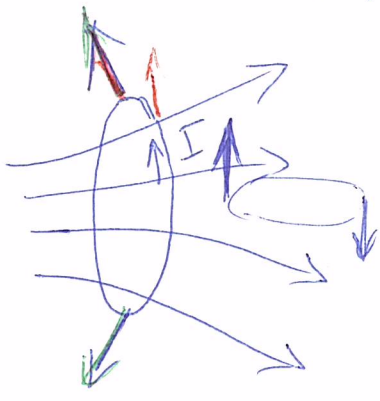
$$\underline{F} = I\,\underline{dl}_j \times \underline{B} \cdot \underbrace{\int_1^2 dr}_{l} = I\,l\,\underline{dl}_j \times \underline{B}$$

for a closed loop in uniform field:

$$\underline{F} = I \oint (\underline{dl}_j \times \underline{B})\,dr = I \underbrace{\oint \underline{dl}_j\,dr}_{\emptyset} \times \underline{B}$$

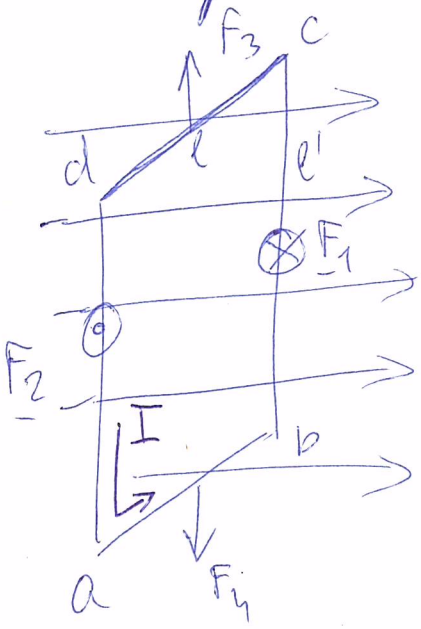
$$\underline{F} = \emptyset$$

In a non-uniform magnetic field?

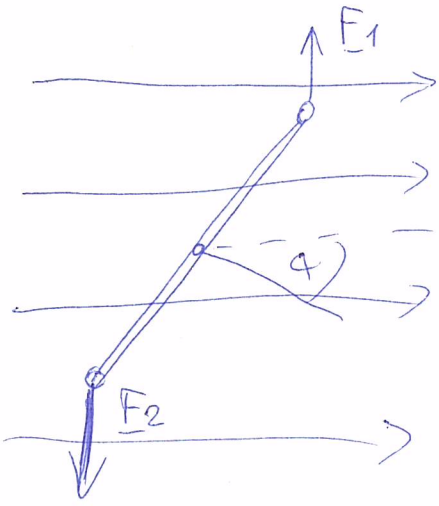


Depending on the current, it might go to a stronger or weaker field direction!

Torque acting on a closed ~~loop~~ circuit:



(Uniform B)



$$F_1 = F_2 = I\, l' B \;\Rightarrow\; M_1 = F_1 \cdot \frac{l}{2} \sin\alpha = M_2$$

$$M_1 + M_2 = M = IB \cdot \underbrace{l l'}_{A} \sin\alpha = I\,|\underline{A} \times \underline{B}|$$

$$\underline{M} = I\,\underline{A} \times \underline{B}$$

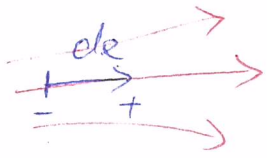
Equilibrium state? $\underline{A} \parallel \underline{B}$

Does it resemble to something we have learnt?

electric dipole

($\underline{M} = \underline{d}_e \times \underline{E}$

equilibrium: $\underline{d}_e \parallel \underline{E}$



)

Very similar! Let's call $I\,\underline{A} = \underline{d}_m$ magnetic dipole-moment

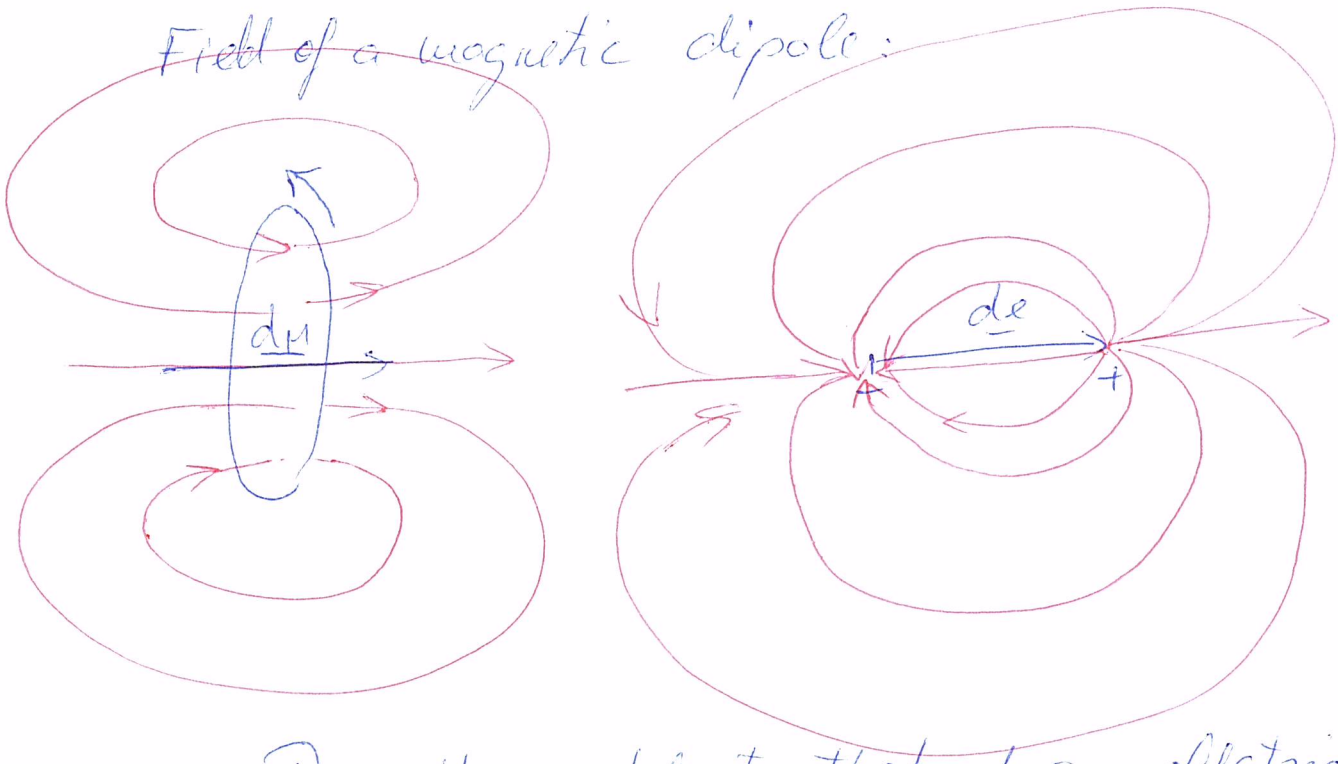


direction: right hand rule

Behaves quite similarly to a compass! Is the needle of the compass a magnetic dipole?

The needle of the compass is a small magnet with N and S poles. But where are the current loops?

Field of a magnetic dipole:

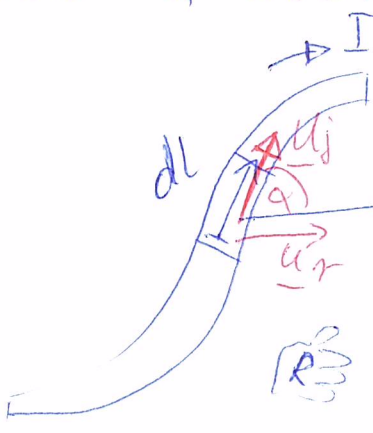


Does it resemble to that of an electric dipole? Not in the ~~AC~~ vicinity of the charges ($r \sim l, R$). Far away? YES

CH 12? Prove that for $r >> l, R$ the field of el. and magnetic dipoles are the same!

It is high time for us to investigate how a current is creating magnetic field!

Biot & Savart (1820) did the experiment (Ampère & Laplace helped the math)



Observations:

$$dB \sim \frac{I\,dl}{r^2}\sin\alpha$$

superposition princ.

$$\underline{dB} \perp \underline{u}_r, \underline{u}_j$$

$$\underline{dB} \parallel \underline{u}_j \times \underline{u}_r$$

$$\Rightarrow \underline{B}(P) = K_m I \oint \frac{\underline{u}_j \times \underline{u}_r}{r^2} dl$$

$K_m := \frac{\mu_0}{4\pi}$ where $\mu_0 \cong 4\pi \cdot 10^{-7} \frac{Vs}{Am}$

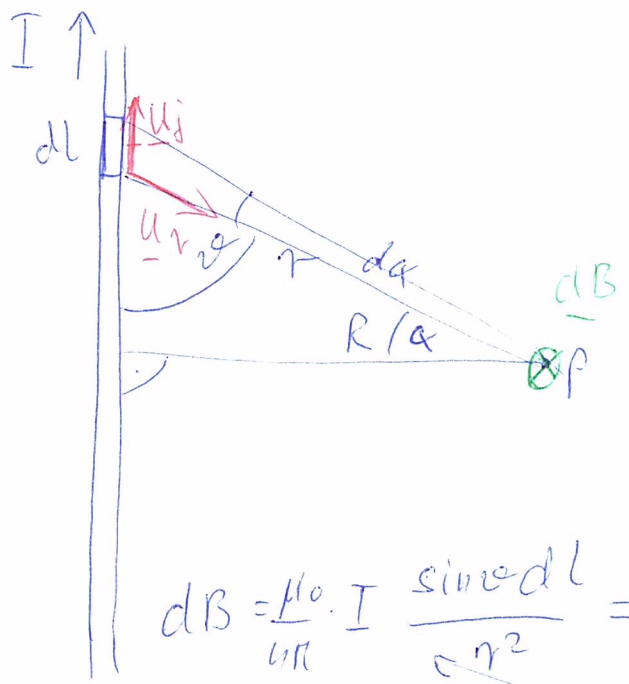
(also redefined in 2019 by a numerical value!)

magnetic PERMEABILITY of vacuum!

(permittivity) for ε_0

Magnetic field of a straight conducting wire

↑ as long

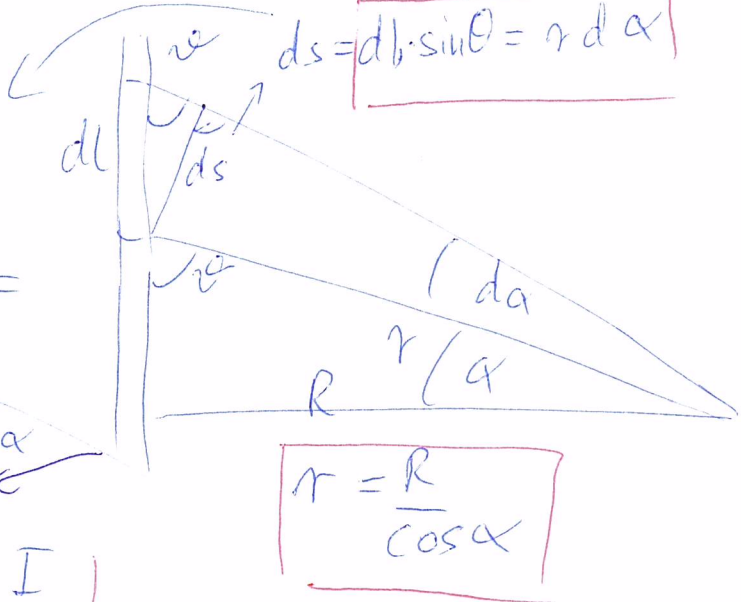


$$d\underline{B} = \frac{\mu_0}{4\pi} I \frac{\underline{u}_j \times \underline{u}_r}{r^2} dl$$

↑ direction: normal to paper, pointing inwards

$$|\underline{u}_j \times \underline{u}_r| = \sin(\pi - \vartheta) = \sin\vartheta$$

$$ds = dl \cdot \sin\vartheta = r\, d\alpha$$



$$r = \frac{R}{\cos\alpha}$$

$$dB = \frac{\mu_0}{4\pi} \cdot I \frac{\sin\vartheta\, dl}{r^2} =$$

$$B = \frac{\mu_0}{4\pi} I \frac{r \cdot d\alpha}{r^2} = \frac{\mu_0}{4\pi} \frac{I \cos\alpha\, d\alpha}{R}$$

$$B = \frac{\mu_0}{4\pi} I \frac{1}{R} \cdot \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\alpha\, d\alpha = \frac{\mu_0 I}{2\pi R}$$

2

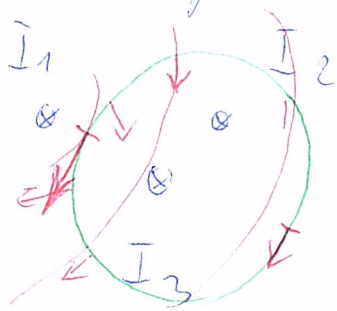
Investigate the rotation of the induction vector around a wire:



$$\oint \underline{B}\, d\underline{r} = \frac{\mu_0 I}{2\pi R} \cdot \int dr = \mu_0 I$$

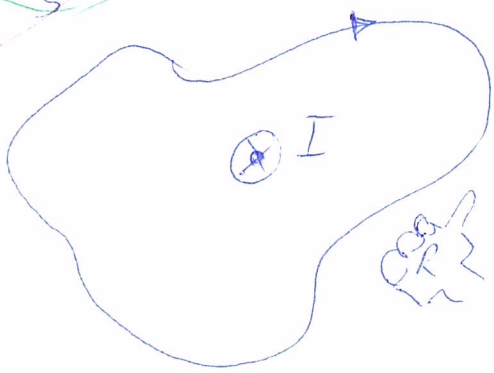
independent of R!

Extrapolate this to more complex geometries:



$\oint \underline{B}_1 d\underline{l} = 0$

$\oint (\underline{B}_2 + \underline{B}_3) d\underline{l} = \mu_0 (I_1 + I_2)$ even if not centered!



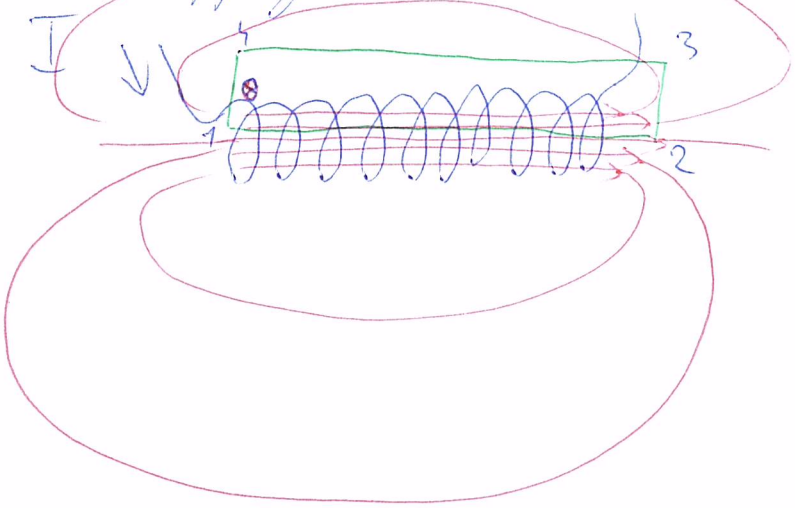
what happens if the closed loop is not a circle?

$$\oint \underline{B}\, d\underline{l} = \mu_0 I$$ (prove?)

Ampère's law $\operatorname{curl} \underline{B} = \mu_0 \underline{j}$

Gauss's law for $\underline{B}$: $\oint\!\!\oint \underline{B}\, d\underline{s} = 0$

Let's apply this to a solenoid coil!



Inside: uniform field

$\int_1^2 \underline{B}\, d\underline{l} = Bl$

$\int_2^3 = 0 \quad \underline{B} \perp d\underline{l}$ $\int_3^4 \underline{B}\, d\underline{l} \approx 0$

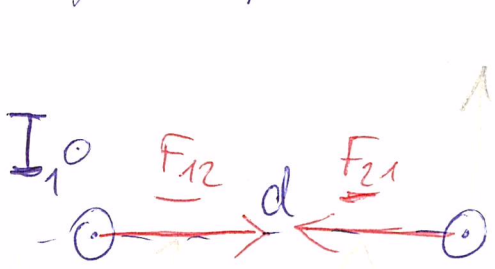
approximation! Works if $l \gg d$ (Ch 12: check!)

$$\oint \underline{B}\, d\underline{l} = Bl = \mu_0 I \cdot N$$

$$B = \frac{\mu_0 I N}{l}$$

Interaction of two straight conductors with current

Magnetic force between || conductors.



$$\underline{F}_{21} = I_2 \, \ell \, \underline{u}_{12} \times \underline{B}_1$$

$$B_1 = \frac{\mu_0 I_1}{2\pi \cdot d}$$

$$F_{21} = \frac{\mu_0 I_1 I_2 \cdot \ell}{2\pi d} = -F_{12}$$

the force for unit length: $f_{21} = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{d}$

Let $I_1 = I_2 = I \Rightarrow$ $\underline{I} = \sqrt{\frac{2\pi f_{21} d}{\mu_0}}$

Original definition of 1A:

if $d = 1m$ $f_{21} = 2 \cdot 10^{-7} \frac{N}{m} \Rightarrow I = 1A$

Has been completely changed in 2019:

$1A = \frac{1\,C}{s}$ (1C defined by the elementary charge e)